

CREATING DATASETS

ds = nc.open_data("foo.nc")
Open a local file as a dataset.

ds = nc.open_url("https://foo.foo.nc")
Open/download a file as a dataset.

ds = nc.open_thredds("https://foo.foo.nc")
Use a thredds/OPeNDAP file as a dataset.

SUBSETTING DATA

ds.subset(lon=[lon_min, lon_max], lat=[lat_min, lat_max])
Crop to a lon/lat box.

ds.subset(variables=[var1, var2])
Select a list of variables.

ds.subset(years=[2000, 2001])
Select a list of years.

ds.subset(months=[5, 6])
Select a list of months.

ds.drop(variables=["var1", "var2"])
Remove a list of variables.

VISUALIZING DATA

ds.plot()
Plot all data in a dataset.

ds.pub_plot()
Create a publication-quality static map.

MERGING METHODS

ds.merge("variable")
Merge a dataset of files with different variables.

ds.merge("time")
Merge a dataset of files with different timesteps.

ROLLING METHODS

Rolling methods require a window to average over.

ds.rolling_mean(20)
Calculate rolling mean using a window of 20.

ds.rolling_min(10)
Calculate rolling min using a window of 10.

ds.rolling_max(5)
Calculate rolling max using a window of 5.

ds.rolling_sum(20)
Calculate rolling sum using a window of 20.

EXPORTING DATASETS

ds.to_xarray()
Export as an xarray dataset.

ds.to_dataframe()
Export as a pandas dataframe.

ds.to_nc("foo/foo.nc")
Export as a netCDF file.

ACCESSING ATTRIBUTES

ds.variables
List dataset variables.

ds.years
List dataset years.

ds.months
List dataset months.

ds.times
List dataset times.

ds.size
Display dataset size.

ds.current
Display dataset files.

TEMPORAL METHODS

Temporal averaging methods take a list specifying the time periods to average over — elements must be "year", "month" or "day". Defaults to an average over all time steps.

ds.tmean("year")
Calculate the annual mean.

ds.tmean(["year", "month"])
Calculate the mean for each month in each year.

ds.tmin()
Calculate the temporal minimum.

ds.tmax()
Calculate the temporal maximum.

ds.tmedian()
Calculate the temporal median.

ds.trange()
Calculate the temporal range.

ds.tpercentile(95)
Calculate the 95th percentile.

ds.tvariance()
Calculate the temporal variance.

ds.shift(hours=-1)
Shift time back 1 hour. Also takes days, months, years.

ds.tcumsum()
Temporal cumulative sum.

ds.first_above(0)
Identify the 1st time step where values are positive.

ds.first_below(0)
Identify the 1st time step where values are negative.

COPYING A DATASET

ds_copy = ds.copy()
Copy a dataset.

VERTICAL METHODS

ds.vertical_mean()
Calculate the vertical mean per grid cell.

ds.vertical_min()
Calculate the vertical minimum per grid cell.

ds.vertical_max()
Calculate the vertical maximum per grid cell.

ds.top()
Extract the top cell, e.g. the sea surface.

ds.bottom()
Extract the bottom cell.

ds.vertical_interp([10, 20, 30])
Interpolate to a list of vertical depths.

ENSEMBLE METHODS

Ensemble methods compare files with the same timesteps and grid. Calculations are done per grid cell.

ds.ensemble_mean()
Calculate the mean across an ensemble.

ds.ensemble_max()
Calculate the maximum across an ensemble.

ds.ensemble_min()
Calculate the minimum across an ensemble.

ds.ensemble_range()
Calculate the range across an ensemble.

GLOBAL SETTINGS

nc.options(lazy=False)
Set evaluation to eager/non-lazy.

nc.options(temp_dir="/foo")
Set the temporary directory to use in this session.

nc.options(cores=6)
Set the number of cores to use when processing multi-file datasets.

nc.options(parallel=True)
Tell NCToolkit multiple datasets will be processed in parallel.

SPATIAL METHODS

Spatial methods are calculated per time step.

ds.spatial_mean()
Calculate the spatial mean.

ds.spatial_min()
Calculate the spatial minimum.

ds.spatial_max()
Calculate the spatial maximum.

ds.spatial_sum()
Calculate the spatial sum.

ds.zonal_mean()
Calculate the zonal mean.

ds.meridional_mean()
Calculate the meridional mean.

ds.to_transect(start=[lon1, lat1], end=[lon2, lat2], nsteps=100)
Extract a straight-line transect between two points.

MULTI-DATASET METHODS

Add, subtract or compare one dataset with another, so long as their grids and timesteps are compatible. Calculations are carried out per timestep and grid cell.

ds + ds1
Add one dataset to another.

ds - ds1
Subtract one dataset from another.

ds * ds1
Multiply a dataset by another.

ds / ds1
Divide a dataset by another.

ds > ds1
Do a dataset's values exceed another's?

ds < ds1
Are a dataset's values less than another's?

RANDOM HACKS

ds.zip()
Zip dataset files.

ds.format("nc4")
Change the netCDF format of dataset files.

ds.as_missing([0, 100])
Set values within a range to missing.

ds.rename({"old_foo": "new_foo"})
Change the name of a variable.

ds.set_units({"var": "foo/s"})
Set the units for a variable.

ds.set_longnames({"foo": "a long foo"})
Set the long name for a variable.

CREATING VARIABLES

New variables can be created using the assign method. This requires a lambda function. Operations are carried out per grid cell and timestep.

ds.assign(new=lambda x: x.old + 10)
Calculate a new variable, which is just an old one plus 10.

ds.assign(new=lambda x: x.old > spatial_mean(x.old))
Create a variable identifying whether a grid cell exceeds the spatial mean.

For more examples, see the [NCToolkit website](https://nctoolkit.github.io).

REGRIDDING

ds.regrid("foo.nc")
Regrid to a file's grid.

ds.regrid(ds2)
Regrid to another dataset's grid.

ds.to_latlon(lon=[lon_min, lon_max], lat=[lat_min, lat_max], res=[lon_res, lat_res])
Regrid to a regular lon/lat grid with a specified extent and resolution.

ds.resample_grid(2)
Resample, selecting every other lon/lat grid cell.